Exploratory Data Analysis of the MIMIC-IV-EXT Triage Instruction Corpus Using Automated EDA Libraries

Contents

1. Introduction
2. Dataset Description
3. Methodology
4. Sweetviz Analysis
5. AutoViz Analysis
6. ydata-profiling Analysis
7. Comparison of Automated EDA Libraries
8. Overall Clinical Insights
9. Conclusion

1. Introduction

Emergency departments handle large volumes of patient data, making timely and accurate clinical decision-making a significant challenge. Exploratory Data Analysis (EDA) is an essential first step in understanding healthcare datasets, as it helps identify data quality issues, summarize feature characteristics, detect missing values, and discover relationships between variables before developing machine learning or artificial intelligence models.

Automated EDA libraries, such as Sweetviz, AutoViz, and ydata-profiling, simplify this process by generating comprehensive reports with statistical summaries, visualizations, feature distributions, correlation analysis, and missing value insights. These tools enable faster and more efficient analysis compared to manual exploration.

In this project, the structured validation dataset from the MIMIC-IV-EXT Triage Instruction Corpus (PhysioNet) was used for exploratory data analysis. The objective of this report is to analyze the dataset using automated EDA libraries, evaluate its quality and characteristics, and identify meaningful clinical insights that can support the future development of a Large Language Model (LLM)-based Clinical Decision Support System (CDSS) for emergency triage.

2. Dataset Description

The dataset used in this project was obtained from the MIMIC-IV-EXT Triage Instruction Corpus, available through PhysioNet. The corpus contains two datasets: the MIETIC dataset with 9,630 instruction, response records for Large Language Model (LLM) training, and the MIETIC-validate-samples dataset containing 50 structured patient records with 57 clinical features.

For this exploratory data analysis, the MIETIC-validate-samples dataset was selected because it provides structured clinical information suitable for automated EDA tools. The dataset includes patient demographics, vital signs, emergency department interventions, medication usage, laboratory events, resource utilization, and expert clinical opinions.

During the initial analysis, the dataset was found to contain 50 rows and 57 columns, with 36 integer, 8 floating-point, 11 categorical, and 2 datetime features. Several clinical variables contained missing values, particularly Expert 3 Opinion, temperature, blood pressure, and other vital sign measurements. No duplicate records were identified. Minimal preprocessing was performed by correcting the first column name, converting the intime and outtime columns to the datetime data type, and saving the prepared dataset for automated exploratory data analysis.

Table 2.1: Dataset Summary

Attribute	Value
Dataset Name	MIETIC-validate-samples
Source	PhysioNet
Number of Records	50
Number of Features	57
Numerical Features	44
Categorical Features	11
Datetime Features	2
Duplicate Rows	0
Missing Values	Present in selected clinical variables
Preprocessing	Column name correction and datetime conversion

Figure 2.1: Dataset preview

	subject_id	stay_id	hadm_id	intime	outtime	gender	race	arrival_transport	disposition	temperature	...	oral_medications	consults_count	p
0	12867223	32373901	25038197.0	2121-01-12 17:47:00	2121-01-12 22:43:00	M	WHITE	AMBULANCE	ADMITTED	NaN	...	2	0	
1	12482649	30078245	24501616.0	2143-03-21 07:46:00	2143-03-21 15:04:00	M	WHITE	AMBULANCE	ADMITTED	97.3	...	0	0	
2	18530850	37186254	25480845.0	2167-02-12 23:52:00	2167-02-13 17:15:00	M	WHITE	AMBULANCE	HOME	96.6	...	4	0	
3	10320289	30019253	27936912.0	2161-06-21 07:53:00	2161-06-21 11:51:00	M	WHITE	AMBULANCE	ADMITTED	NaN	...	1	0	

Figure 2.2: Missing value summary table

	Missing Count	Missing Percentage
Expert 3 Opinion	34	68.0
temperature	17	34.0
dbp	14	28.0
resprate	14	28.0
sbp	13	26.0
o2sat	13	26.0
heartrate	12	24.0
hadm_id	10	20.0
pain	5	10.0

3. Methodology

The exploratory data analysis (EDA) was conducted using a structured workflow to understand the clinical dataset before developing machine learning and Large Language Model (LLM)-based solutions. First, the structured validation dataset was loaded into Pandas for initial exploration. The dataset was examined to identify its dimensions, data types, missing values, duplicate records, and descriptive statistics.

Next, minimal preprocessing was performed by correcting the first column name, converting the intime and outtime columns to the datetime format, and saving the prepared dataset. No extensive data cleaning or feature engineering was performed, allowing the automated EDA tools to analyze the original characteristics of the data.

The prepared dataset was then analyzed using three automated EDA libraries. **Sweetviz** was used to generate an interactive HTML report containing dataset summaries, feature distributions, missing value analysis, and feature associations. **AutoViz** was used to automatically create visualizations for numerical and categorical variables, while **ydata-profiling** generated a comprehensive data profiling report, including data quality checks, correlations, interactions, and statistical summaries. Finally, the outputs from these tools were interpreted and compared to identify meaningful clinical insights and evaluate their usefulness for emergency triage data analysis.

4. Sweetviz Analysis

Sweetviz is an automated exploratory data analysis (EDA) library that generates interactive HTML reports with minimal coding effort. It provides a comprehensive overview of a dataset by summarizing feature distributions, missing values, statistical information, and relationships between variables. These reports help analysts quickly identify data quality issues, understand feature characteristics, and discover patterns within the data.

For this project, Sweetviz was applied to the prepared clinical dataset to examine patient demographics, vital signs, emergency department interventions, medication usage, and expert clinical annotations. The generated report was analyzed to gain insights into the dataset and identify characteristics that may support the development of an AI-based Clinical Decision Support System (CDSS) for emergency triage.

4.1 Sweetviz Report Generation

The prepared dataset was analyzed using the Sweetviz library in Python. An automated HTML report was generated to summarize the dataset and visualize important characteristics such as feature distributions, missing values, and associations between variables. This report served as the basis for the exploratory data analysis presented in the following sections.

Figure 4.1: Python code used to generate the Sweetviz HTML report.

```
# Generate the Sweetviz report
report = sv.analyze(df)

# Save the report as HTML
report.show_html(
    "Sweetviz_Report.html",
    open_browser=True
)
```

4.2 Dataset Overview

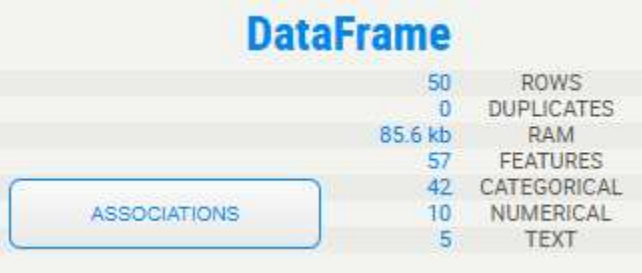
The Sweetviz report provides a comprehensive overview of the prepared clinical dataset before presenting detailed feature-level analysis. It confirms that the dataset contains **50 patient records** and **57 features**, with **no duplicate records**, indicating good data quality. Based on Sweetviz's automatic classification, the dataset consists of **42 categorical features, 10 numerical features, and 5 text features**.

This overview helps in understanding the overall structure and composition of the dataset before examining individual variables. The dataset includes patient demographics, vital signs, emergency department interventions, medication usage, resource utilization, and expert clinical opinions. The presence of different feature types allows for a comprehensive exploratory analysis of the clinical data.

Key Observations

- The dataset contains **50 patient records** and **57 features**.
- No duplicate records were identified.
- Sweetviz automatically classified the features into **42 categorical**, **10 numerical**, and **5 text** variables.
- The dataset contains a combination of demographic, clinical, treatment, and expert annotation features.
- The structured nature of the dataset makes it suitable for automated exploratory data analysis.

Figure 4.2: Sweetviz dataset overview



4.3 Feature Summary

The Sweetviz report provides a summary of each feature in the dataset, including descriptive statistics, missing values, unique values, and data distributions. It helps identify the characteristics of both numerical and categorical variables, making it easier to understand the overall quality and structure of the dataset.

The report shows that the dataset contains a combination of patient demographic information, vital signs, clinical interventions, medication usage, and expert clinical opinions. These feature summaries provide a strong foundation for identifying important variables relevant to emergency triage.

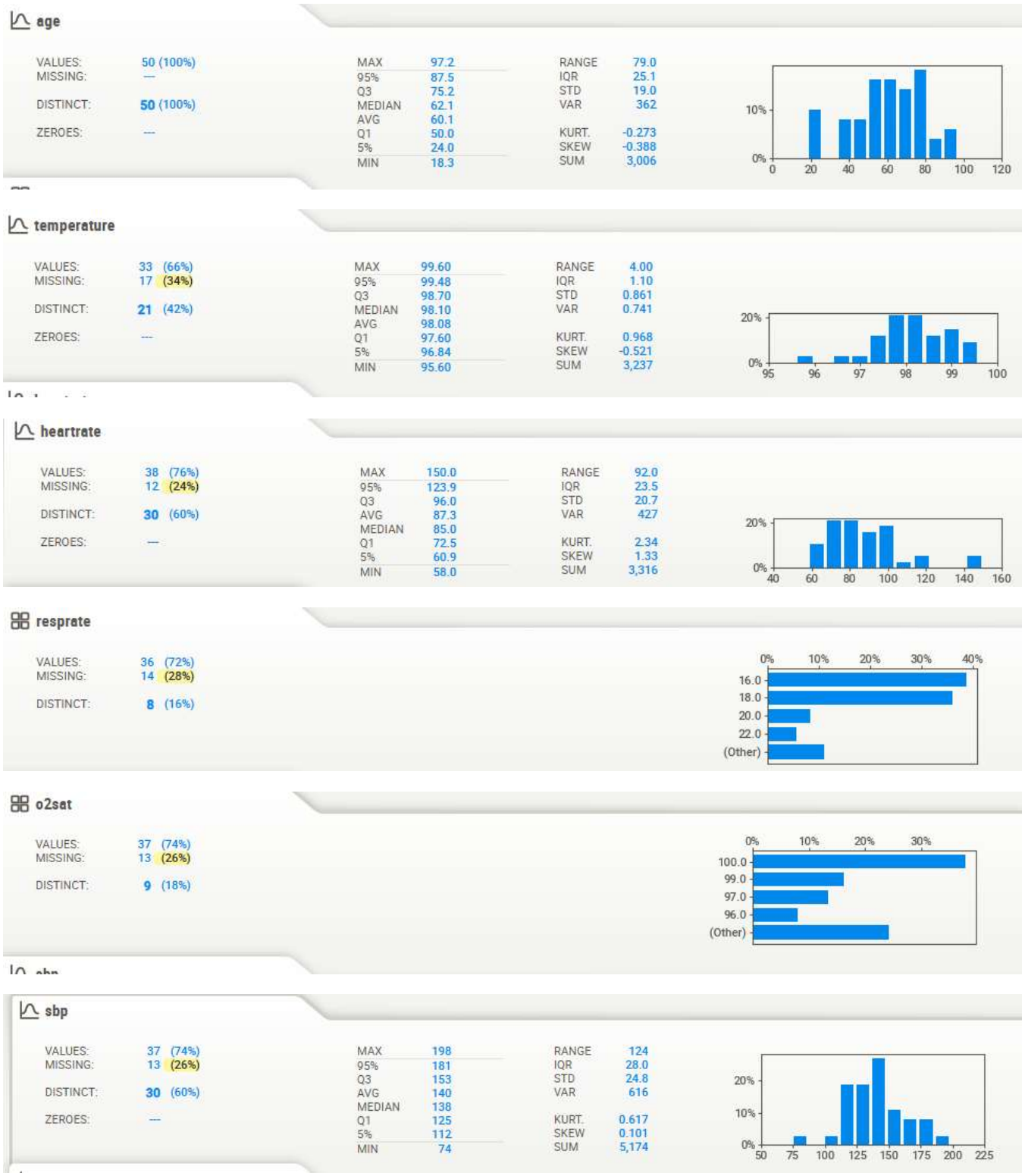
Key Observations

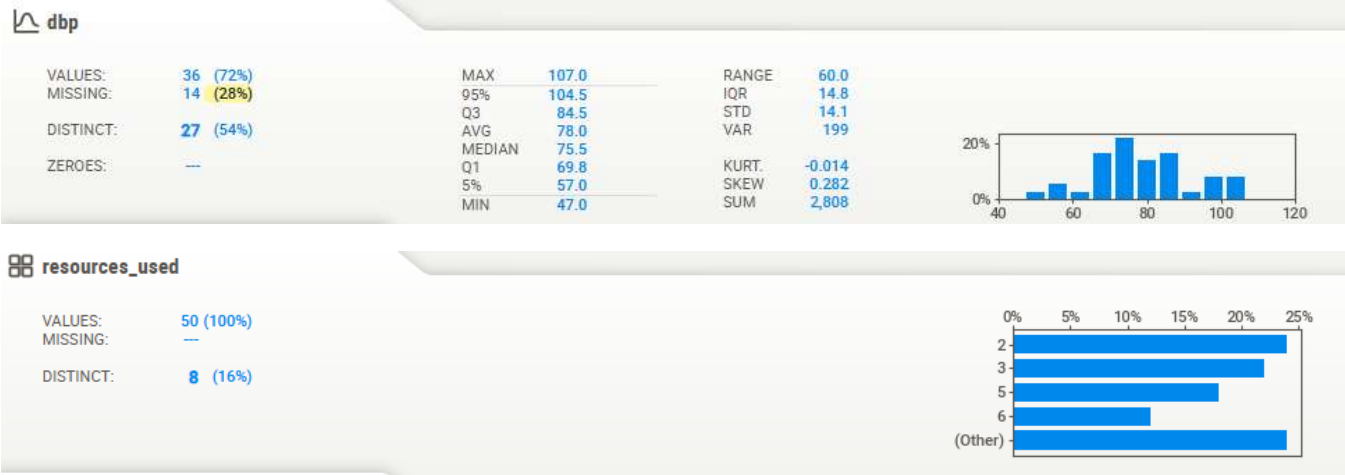
- Sweetviz summarizes every feature using statistical and graphical information.
- Numerical features include descriptive statistics and distribution plots.
- Categorical features display unique values and frequency distributions.
- The dataset contains clinically relevant variables for emergency triage analysis.

4.4 Numerical Feature Analysis

The Sweetviz report provides statistical summaries and distribution plots for all numerical variables in the dataset, including age, vital signs, medication counts, procedure counts, and resource utilization. These summaries help understand the distribution, variability, and completeness of clinical measurements, which are important for emergency triage analysis.

Figure 4.4: Statistical summaries and distributions of selected numerical features generated by Sweetviz.



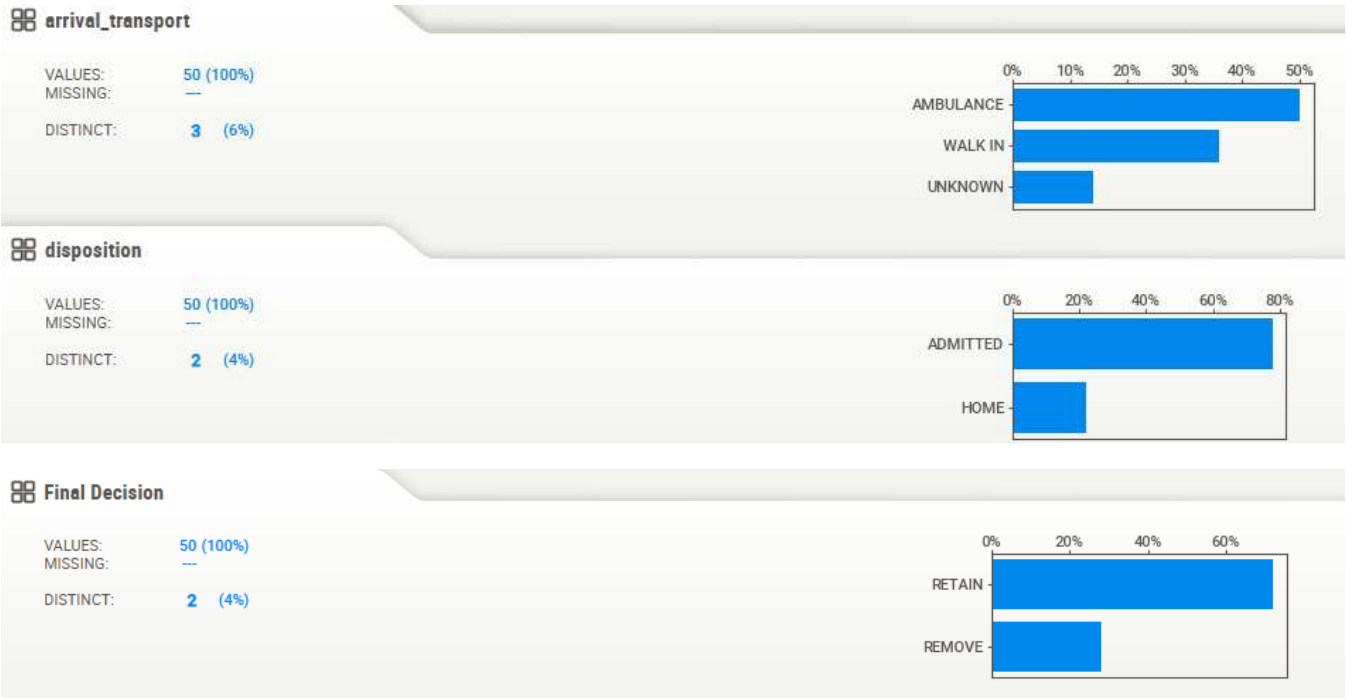


4.5 Categorical Feature Analysis

The Sweetviz report summarizes the distribution of important categorical variables, including patient gender, race, arrival transport, disposition, and final decision. These variables provide valuable insights into patient demographics, methods of arrival, clinical outcomes, and expert review decisions. The distributions help identify common patient characteristics and support understanding of the emergency department population.

Figure 4.4: Statistical summaries and distributions of selected categorical features generated by Sweetviz.





4.6 Missing Value Analysis

The Sweetviz report highlights the missing values present in the dataset, helping identify variables that may require preprocessing before model development. Several clinical measurements contain missing values, while most demographic and intervention-related variables are complete.

The highest percentage of missing values was observed in **Expert 3 Opinion (68%)**, followed by **temperature (34%)**, **diastolic blood pressure (DBP) (28%)**, **respiratory rate (28%)**, **systolic blood pressure (SBP) (26%)**, **oxygen saturation (26%)**, **heart rate (24%)**, **hadm_id (20%)**, and **pain (10%)**.

4.7 Feature Associations

The Sweetviz report provides feature association analysis to identify relationships between variables. Understanding these associations helps determine which features may influence one another and which variables could be useful predictors in future machine learning models.

The association analysis assists in identifying clinically relevant relationships among patient demographics, vital signs, emergency interventions, and treatment outcomes. These insights are valuable for feature selection and predictive modeling in emergency triage applications.

4.8 Clinical Insights

The Sweetviz analysis provided valuable insights into the characteristics and quality of the emergency triage dataset. The dataset contains diverse clinical information, including patient demographics, vital signs, emergency department interventions, medication usage, and expert clinical opinions. While most variables are complete, some vital signs and expert annotations contain missing values that should be addressed during data preprocessing.

The exploratory analysis indicates that the dataset is suitable for developing AI-based clinical decision support systems. Variables such as **age, vital signs, disposition, resource utilization, and expert decisions** may serve as important predictors for emergency triage and risk assessment.

5. AutoViz Analysis

5.1 Introduction to AutoViz

AutoViz is a highly efficient automated Exploratory Data Analysis (EDA) library in Python that rapidly generates comprehensive visualizations for any dataset with a single line of code. It automatically identifies data types, selects the most appropriate chart types, and maps the relationships between features and a specified target variable. For this project, AutoViz was used to visually diagnose the distribution of clinical variables and their relationship to triage severity (Final Decision).

5.2 Visualization generated

· **Histograms:** Used to assess the distribution and skewness of continuous numerical features, such as patient age and vital signs (temperature, heart rate, respiratory rate).

· **Boxplots:** Generated to identify statistical outliers within the clinical measurements and to observe how vital signs fluctuate across different triage categories.

· **Scatter Plots:** Utilized to map pairwise relationships between continuous variables, highlighting potential linear correlations.

· **Heatmaps:** Provided a high-level color-coded matrix of correlation coefficients across all numerical data points.

5.3 Observations & Findings

· **Distributions:** The histograms indicate that while some vital signs approach a normal distribution, variables such as resource_used and certain medication usages are highly skewed, which is typical for emergency department data where only a subset of patients require extensive interventions.

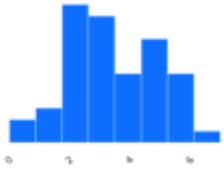


Fig1: resource_used

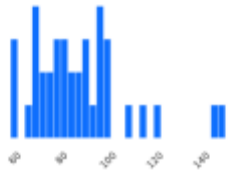


Fig 2: heartrate

· **Outliers via Boxplots:** Several boxplots highlighted extreme outliers in blood pressure (sbp, dbp) and heart rate. These outliers are critical as they often represent the most severe, high-acuity patients requiring immediate life-saving interventions.

· **Target Relationships:** When grouping the data by the target variable (Final Decision), the visual spread showed clear distinctions. Patients categorized into more severe triage levels visually demonstrated tighter clusters of abnormal vital signs (e.g., lower o2sat and higher resprate).

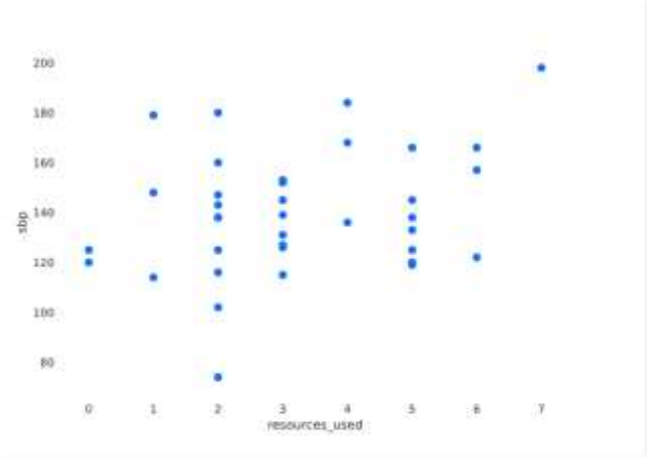


Fig 3: interaction between resource_used and sbp

6. ydata-profiling Analysis

The ydata-profiling library was utilized to generate a comprehensive, interactive HTML report serving as a deep statistical audit of the dataset. The analysis confirmed the following high-level statistics for the MIETIC-validate-samples dataset:

- **Total Observations (Rows):** 50
- **Total Variables (Columns):** 57
- **Missing Cells:** 132 (4.6%)
- **Duplicate Rows:** 0 (0.0%)



1. Alerts & Data Quality Warnings

The profiling tool generated several critical alerts that must be addressed prior to machine learning modelling

- **High Cardinality:** Text-based columns such as chiefcomplaint, subject_id and tiragecase are 100% unique. Patient identifier columns (stay_id, hadm_id) also possess high cardinality and offer no predictive value.
- **Constant / Zero Variance:** A significant number of clinical intervention flags contain only zeros (e.g., expired_within_1h, transfer_within_1h, transfer_to_icu_in_1h). Because these values never change within this 50-row sample, they provide zero variance.
- **Imbalance:** Features like invasive_ventilation and psychotropic_med_within_120min are highly imbalanced, consisting of 98% to 100% zeros.
- **Missing Data (4.6% overall):** Missingness is concentrated within specific clinical parameters (such as optional vital signs or secondary laboratory counts), indicating standard real-world data entry gaps in an emergency department.

2. **Missing value assessment**

The overall missing cell rate of 4.6% indicates that while the dataset is mostly complete, it reflects the non-mandatory nature of certain clinical measurements during emergency triage. In a clinical decision support context, a 4.6% missingness rate means that any downstream predictive model cannot safely assume all vital signs will be present. A robust imputation strategy (such as using the median value of a feature or K-Nearest Neighbors imputation) must be incorporated into the preprocessing pipeline to handle these gaps without losing valuable patient records.

3. **Correlation**

The profiling heatmap identified clear instances of columns giving the same information:

- sbp (Systolic Blood Pressure) is highly correlated with dbp (Diastolic Blood Pressure).
- heartrate is strongly correlated with resprate.
- resources_used shows a high correlation with the patient's final acuity level and clinical disposition.

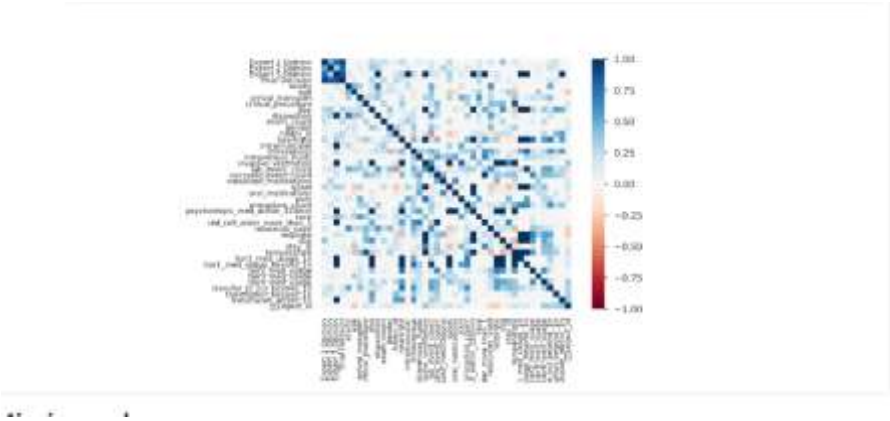


Fig 4:correlation heatmap

4. **Observations & Recommendations**

- Any column where every patient has the exact same answer (like a column filled entirely with zeros) needs to be deleted because it doesn't give the AI any useful information.
- Since 4.6% of the medical numbers (like vital signs) are missing, we need to use a mathematical method to fill in those empty spots so the model doesn't crash
- The chiefcomplaint column contains unique text notes for every patient. We need to use text-processing tools (NLP) to convert these written words into a format the computer can understand alongside the regular numbers.

7. Comparison of Automated EDA Libraries

The three automated EDA libraries used in this project, **Sweetviz, AutoViz, and ydata-profiling** provided complementary insights into the clinical dataset.

Sweetviz offered an interactive overview of the dataset with detailed feature summaries, missing value analysis, and feature associations. Its user-friendly interface made it easy to explore individual variables and understand the overall structure of the dataset.

AutoViz focused on automatic visualization by generating histograms, boxplots, scatter plots, and heatmaps. These visualizations helped identify data distributions, outliers, and relationships between clinical variables, making it particularly useful for understanding patient characteristics and potential predictors.

ydata-profiling produced a comprehensive statistical report by identifying missing values, high-cardinality features, constant variables, imbalanced columns, and correlations. It also generated data quality alerts that are valuable for preprocessing before machine learning model development.

Overall, each tool contributed unique strengths. Sweetviz provided detailed feature exploration, AutoViz emphasized visual interpretation, and ydata-profiling focused on data quality assessment. Using these tools together resulted in a more complete understanding of the emergency triage dataset.

8. Overall Clinical Insights

The exploratory data analysis provided valuable insights into the emergency triage dataset and confirmed that it is suitable for developing an AI-based Clinical Decision Support System (CDSS). The dataset contains a combination of patient demographics, vital signs, emergency department interventions, medication usage, resource utilization, and expert clinical opinions.

The analysis showed that most patients in the sample were middle-aged or older adults, with complete demographic information and no duplicate records. Several important clinical variables, including temperature, heart rate, respiratory rate, oxygen saturation, and blood pressure, contained missing values that should be handled during data preprocessing. The reports also identified highly imbalanced and constant variables, as well as high-cardinality text features that require appropriate preprocessing techniques before model development.

Correlation and visualization analyses indicated relationships among several clinical measurements, suggesting that variables such as age, vital signs, resource utilization, and clinical disposition may serve as important predictors for emergency triage decisions. Overall, the findings demonstrate that the dataset provides meaningful clinical information while highlighting the preprocessing steps necessary for reliable machine learning and LLM-based applications.

9. Conclusion

This project performed an exploratory data analysis of the MIMIC-IV-EXT Triage Instruction Corpus structured validation dataset using three automated EDA libraries: Sweetviz, AutoViz, and ydata-profiling. The analysis provided a comprehensive understanding of the dataset by examining its structure, feature distributions, missing values, correlations, and overall data quality.

Each EDA tool contributed different strengths to the analysis. Sweetviz offered detailed feature-level exploration, AutoViz generated informative visualizations, and ydata-profiling performed an extensive statistical and data quality assessment. Together, these tools helped identify important clinical characteristics, potential predictive variables, and data quality issues that should be addressed before machine learning model development.

The results indicate that the dataset is a suitable foundation for future work on AI-powered emergency triage and Clinical Decision Support Systems. Future work will involve data preprocessing, feature engineering, and the development of Large Language Model (LLM)-based models to support emergency department decision-making.